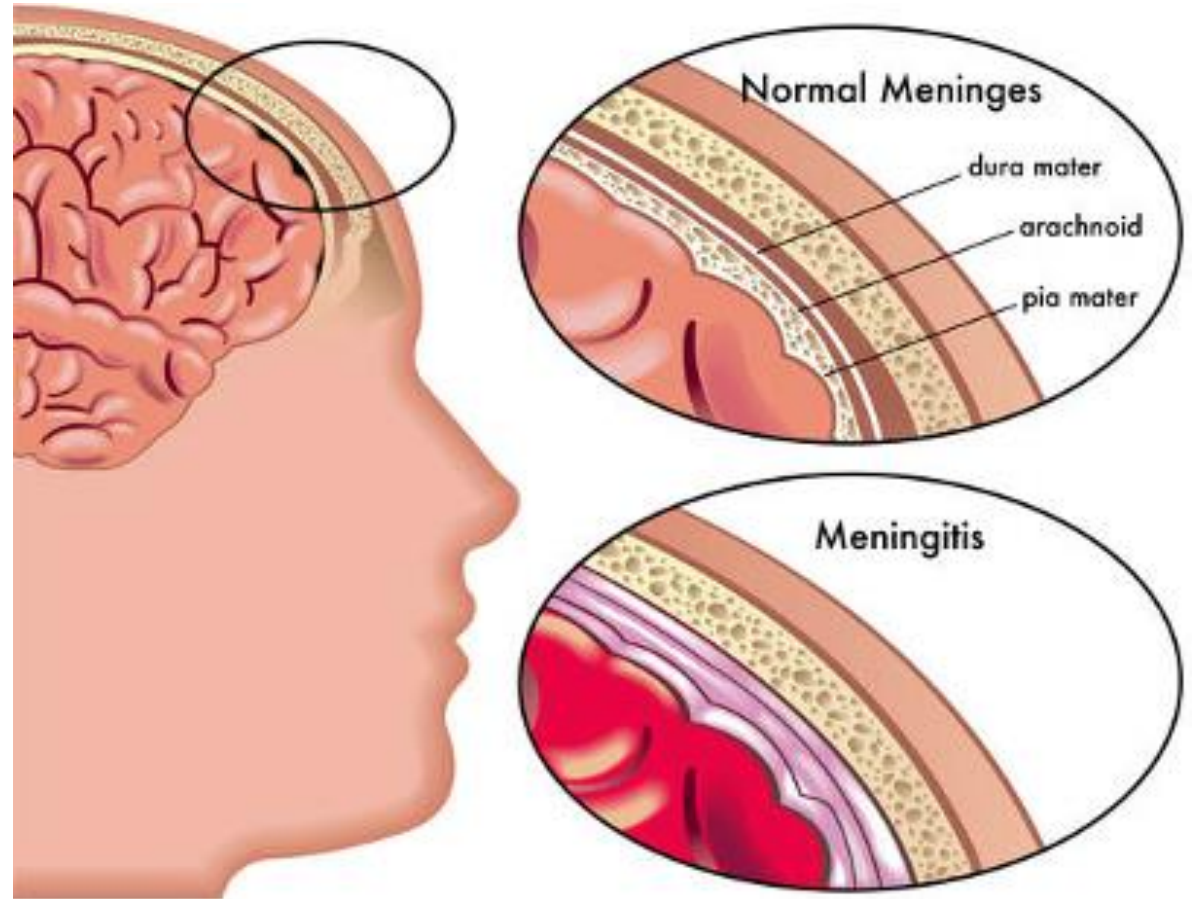


Bacterial meningitis

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Outlines

- Definition of the meningitis
- Pathophysiology of the meningitis
- Risk factors of the meningitis
- Clinical Presentation of the meningitis
- Complications of the meningitis
- Diagnostic Tests of the meningitis
- Management of the meningitis

Pre test

Mention causes of
bacterial meningitis

Mention
diagnostics test for
bacterial meningitis

Definition

Meningitis is a life-threatening disorder that is most often caused by bacteria or viruses.



Bacterial meningitis is an inflammation that involves the arachnoid and pia mater of the brain, the subarachnoid space, and the CSF.

Causes of bacterial meningitis



Meningitis can be caused by infectious and non-infectious processes (autoimmune disorders, cancer/paraneoplastic syndromes, drug reactions).



The infectious etiologic agents of meningitis include bacteria, viruses, fungi, and less commonly parasites.

Risk Factors

- **Risk factors for meningitis include:**

- Chronic medical disorders (renal failure, diabetes, adrenal insufficiency, cystic fibrosis)
- Extremes of age
- Under vaccination
- Immunosuppressed states (iatrogenic, transplant recipients, congenital immunodeficiencies, AIDS)
- Living in crowded conditions
- Alcohol use disorder
- Presence of ventriculoperitoneal (VP) shunt
- Presence of external ventricular drain
- Penetrating head injury
- Lumbar puncture
- Cranial or spinal surgeries involving the opening of the dura
- Bacterial endocarditis

Pathophysiology

- Bacteria enter the CNS through the bloodstream and cross the blood-brain barrier or directly enter the bloodstream via penetrating trauma, invasive procedures, cancer, certain drugs, or ruptured cerebral abscess.

An upper respiratory infection → bacteria in nasopharynx → bloodstream → CSF subarachnoid space and pia arachnoid membrane → meninges.

Purulent exudate → clings to meningeal layers → clogs CSF → tissue and vascular congestion and obstruction → cranial nerve dysfunction, hyperemia of meningeal blood vessels, brain tissue edema, ↑ CSF, ↑ ICP, ↑ WBC in subarachnoid space → acute hydrocephalus and seizures.

Abnormal stimulation of hypothalamic area → inappropriate ADH production → water retention.

Clinical Presentation

■ Symptoms of upper respiratory infection possibly preceding meningeal irritation →

severe and unrelenting headache,

nausea,

vomiting,

fever and chills, nuchal rigidity (stiff neck),

irritability, malaise, restlessness,

myalgia, and tachycardia

Photophobia and signs of ↑ ICP

■ Problems with memory; ↓ LOC; disorientation to person, place, and year;

sleepiness,

confusion,

irritability,

abnormal eye movements → coma, delirium, and seizures

■ Kernig's sign: inability to extend the leg at the knee when the thigh is flexed

■ Brudzinski's sign: flexion of the hip and knee when the patient's neck is flexed

Complications

-
- Septic emboli and septic shock with vascular dysfunction, or DIC
-
- Pericardial effusion
-
- Fluid and electrolyte imbalances
-
- Seizures and hemiparesis
-
- Cranial nerve (CN) dysfunction: CN III, IV, VI, VII, VIII
-
- Cerebral infarction, brain parenchymal damage
-
- Hydrocephalus and cerebral edema
-
- ↑ ICP → herniation of the brain

Diagnostic Tests



■ Lumbar puncture, culture, and assessment of CSF and pressure → ↑ CSF pressure (CSF pressure >180 mm Hg is indicative of bacterial meningitis);



CSF fluid cloudy with ↑ protein, ↓ glucose, ↑ WBCs and neutrophils



■ CBC, especially ↑ WBC



■ Blood cultures and cultures from nasopharynx, respiratory secretions, urine



■ Serum electrolytes, especially Na (dilutional hyponatremia), BUN, glucose



■ CXR to determine concomitant pneumonia



■ CT scan or MRI if ↑ ICP, or brain abscess or hydrocephalus

Medical Management

Antibiotics and supportive care are critical in all cases of bacterial meningitis.

Managing the airway, maintaining oxygenation,

Giving sufficient intravenous fluids while providing fever control are parts of the foundation of meningitis management.

The type of antibiotic is based on the presumed organism causing the infection.

The clinician must take into account patient demographics and past medical history in order to provide the best antimicrobial coverage.

Current Empiric Therapy

Adults (18 to 49 years old)

Ceftriaxone IV and

Vancomycin IV

Adults older than 50 years old and the immunocompromised

Ceftriaxone IV and

Vancomycin IV and

Ampicillin IV

Steroid Therapy

There is insufficient evidence to support the widespread use of steroids in bacterial meningitis.

Some studies report a reduction in mortality for *Streptococcus pneumoniae* meningitis, but not in *Haemophilus influenza* or *Neisseria meningitidis* meningitis.

Increased Intracranial Pressure

If the patient develops clinical signs of increased intracranial pressure (altered mental status, neurologic deficits, non-reactive pupils, bradycardia), interventions to maintain cerebral perfusion include:

Elevating the head of the bed to 30 degrees

Inducing mild hyperventilation in the intubated patient

Osmotic diuretics such as 25% mannitol or 3% saline

Nursing Management

Monitoring and recording vital signs.

Assess the Glasgow Coma Scale and pay special attention to the motor response, immediately inform the on-duty doctor if it falls.

Assess the patient's mental status and provide psychological support if the patient is conscious.

Elevate the head of the bed to 30 degrees with a straight neck for venous drainage from the brain.

Ensure the patient has an IV line for fluids and medications.

Administer antibiotics as prescribed

Nursing Management

- Maintain respiratory isolation until pathogen is not cultured in nasopharynx (usually 24 hr after antibiotic treatment).

- Monitor neurological status, cranial nerve function, and vital signs every 1–2 hr.

Check pupils, LOC, and motor activity.

- Assess vascular function for signs of septic emboli.

- Ensure ↓ environmental stimuli, quiet environment, and ↓ exposure to lights.

- Administer corticosteroids to decrease inflammation: dexamethasone.

- Administer anticonvulsants for seizures: phenytoin, phenobarbital, lorazepam.

-Administer antipyretics for fever.

-Administer analgesia for headache.

-Administer hyperosmolar agents for cerebral edema: mannitol and diuretics: Lasix.

-Insert surgical shunt if hydrocephalus is present and adjust antibiotic therapy per the causative agent and antibiotic sensitivity.

-Consider the following antibiotic therapy:

-Cefotaxime (Claforan) ■ Ceftazidime (Ceptaz, Fortaz)

-Ceftriaxone (Rocephin)

-Vancomycin

-Meropenem (Merrem) Assess CSF analysis, Gram stain, and cultures for antibiotic sensitivity. Seizure Disorder

-Assess hearing power in both ears.

-Check labs (meningitis may cause SIADH)

-Check renal and liver function tests.

-Check cerebrospinal fluid lab results.

-Change position every 2-hourly to prevent bed sore if the patient is bedridden and apply air mattress and thromboembolic deterrent stockings as needed)

-Provide limb physio and chest physio to prevent the complications associated with immobility.

Never forget to measure random blood sugar before lumbar puncture while sending CSF for a glucose test.

A proper diet chart to provide adequate calories required for the body.

Put a nasogastric tube and start feeding for the unconscious patients, as instructed

Monitor intake and output.

Provide comfort measures

Nursing Diagnosis

Ineffective tissue perfusion related to cerebral edema as evidenced by hypercapnia.

Hyperthermia related to infection as evidenced by body temperature above the normal range.

Acute pain related to increased intracranial pressure as evidenced by headache.

Disturbed sensory perception related to decreased loss of consciousness as evidenced by altered sensorium.

Anxiety-related to change in the health status of a patient as evidenced by expressed concern and worry about actual hospitalization and seriousness of the illness.

Deficient knowledge related to lack of exposure to information as evidenced by request for information about medications, signs and symptoms, and behaviors to report.

The risk for injury related to the internal factors of altered neurologic regulatory function.

Impaired physical mobility related to intravenous infusion, nuchal rigidity, and restraining devices.

Interrupted family processes related to the critical nature of the situation and uncertain prognosis.

Ineffective airway clearance related to neuromuscular damage as evidenced by difficulty in breathing.



POST TEST

Mention causes
of bacterial
meningitis

Mention
complication of
bacterial
meningitis



Any question

Thank you
